Advertisement Intelligence and Persuasion Analytics System

Applied to the ManiPsych repository

Audit, integration, and challenge-round report

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Corpus: 5,738 ads · 5,717 annotations · 642 known same-source links

Taxonomy: adintel-taxonomy-v2

Verdict: PASSED WITH DOCUMENTED RISKS

This document is the final deliverable of the Advertisement Intelligence and Persuasion Analytics System application. It contains 21 sections: an executive summary, the current architecture and data-flow map, baseline results, audit findings, the revised technique ontology, the research and evidence ledger, checkpoint comparison, persuasive-profile design, clustering analysis, authorship and common-source analysis, outlier and novelty analysis, performance and causal-evidence analysis, root-cause analyses, the implementation ledger, two challenge-round defect ledgers, final build evidence, the experiment backlog, unresolved risks, an owner-facing value summary, and the final checkpoint verdict.

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1. Executive Summary

This report documents the application of the Advertisement Intelligence and Persuasion Analytics System specification to the existing ManiPsych repository. The repository is a defensive research pipeline for detecting psychological-manipulation techniques in Peruvian 'ayuda económica' classified ads. The original system had a 20-label flat taxonomy, a TF-IDF one-vs-rest logistic-regression council model, an annotation GUI, and an HTML observatory dashboard. The new specification demands a substantially more rigorous system: a hierarchical multi-label taxonomy, a 17-dimension persuasive-language profile, seven cluster spaces with stability evaluation, four separate authorship tasks with length-aware abstention, ten outlier types with full provenance, a checkpoint registry with calibration and abstention, two adversarial challenge rounds, and twenty-one final deliverables.

We implemented the new specification as a Python package adintel/ with nine modules: types.py (typed dataclasses for every output), taxonomy.py (hierarchical taxonomy v2 with v1-to-v2 mapping), profile.py (17-dimension persuasive profile), clustering.py (seven-space clustering with stability and leakage evaluation), authorship.py (pairwise verification, closed-set, open-set, creative-source clustering), outlier.py (ten outlier detectors), checkpoints.py (registry, calibration helpers, no-averaging-uncalibrated rule), api.py (JSON API with versioned typed outputs), and evidence.py (causal-language linting, universal-score guard, identity guard). The package was built test-first: 113 new tests were written before or alongside implementation, all passing.

The pipeline was run end-to-end on the real corpus of 5,738 ads. Authorship verification on 50 known same-source pairs achieved 100.0% accuracy with 0 length-aware abstention. Outlier detection on a 1,000-ad sample produced 795 reports across 7 types. Clustering ran in all seven spaces with stability ARI ranging from 0.42 to 1.0. Two adversarial challenge rounds identified 18 defects (4 critical/high fixed in-session, 7 medium/low fixed in-session, 7 deferred with documented limitations).

The final verdict is **PASSED WITH DOCUMENTED RISKS**. The system meets every spec requirement at the implementation level, but four risks prevent promotion to full PASSED: (1) labels remain weak-supervised (council suggestions, not human-adjudicated gold); (2) the corpus has no image pixels, blocking visual-persuasion modelling; (3) authorship thresholds are calibrated on positive-only pairs (no negatives), so false-positive rate is unknown; (4) the annotation GUI still uses v1 labels (v2 migration script exists but the GUI generator needs UX work to render the hierarchy). These risks are documented in §19 and the experiment backlog in §18.

2. Current Architecture and Data-Flow Map

The ManiPsych repository is organised as a local-first Python project. The original pipeline runs in six phases: Phase 1 builds a manipulation-technique compendium; Phase 2 builds a Peru sociological dossier; Phase 3 researches forum discussion of 'ayuda económica' ads; Phase 4 collects public raw HTML from Doplim, Locanto, Evisos, Facebook, and Ciudad Anuncios; Phase 5 trains detection models; Phase 6 integrates everything into an HTML report and annotation GUI. Each phase has a machine-checkable gate in tools/phase_gate.py.

The data flow is: scrapers (tools/collect_*.py) write raw HTML to data/raw/ads/; tools/rebuild_manifest_from_raw.py produces data/processed/ad_manifest.jsonl with PII-redacted title and body; tools/prepare_annotation_campaign.py freezes the corpus and generates annotation assignments; tools/run_council_annotation_pass.py runs a three-subagent council that produces suggestion-layer annotations stored in data/annotation/; tools/council_consensus.py resolves council agreement at the 90% threshold; tools/train_manipulation_model.py and tools/train_council_candidate_model.py train TF-IDF OVR logistic-regression baselines; tools/render_html_report.py and tools/generate_council_inferences_report.py produce the HTML observatory and ranking JSON; tools/generate_annotation_gui.py produces the in-browser annotation studio.

The new adintel/ package slots in alongside the existing pipeline without modifying it. The pipeline runner scripts/run_adintel_pipeline.py reads the same manifest and council-annotations files, runs the persuasive profile, seven-space clustering, authorship verification on known same-source pairs, and outlier detection, and writes results to reports/adintel/. The dashboard generator scripts/generate_adintel_dashboard.py renders those results as a self-contained HTML audit page. The migration script scripts/migrate_v1_annotations_to_v2.py projects the existing 5,717 annotations forward to the v2 taxonomy without destroying the v1 labels. All existing v1 tests continue to pass; no v1 behaviour was changed.

3. Baseline Test and Model Results

Before integration, the repository had 75 tests, of which 74 passed and 1 failed. The single failure was environmental: the phase-gate blackbox test requires every raw HTML file referenced by the manifest to exist on disk, but only a sample of the 2 GB raw archive was uploaded. This is not a code defect. We also fixed one portability bug: tools/collect_hombre_locanto.py had a hardcoded macOS temp path (/var/folders/...) that broke on Linux; we replaced it with a MANIPSYCH_SCRATCH environment variable defaulting to ROOT/SCRATCH/implementer. After this fix, 74 of 75 baseline tests passed.

The baseline council model (models/manipulation_council_tfidf_ovr.joblib) was trained on 5,717 weakly-labelled records. Its held-out metrics are: test micro-F1 0.9008, macro-F1 0.7044, subset accuracy 0.5076, label accuracy 0.9610, micro ROC-AUC 0.9872, supported macro ROC-AUC 0.9721. These are agreement-with-council metrics, not independent human-validity metrics, because the council labels are themselves weak-supervised suggestion data (gold=false). The model card in reports/model_card.md explicitly discloses this limitation.

The label distribution is highly skewed. reciprocity_obligation appears on 5,711 of 5,717 records (99.9%) because the v1 labelling rule fired on every 'ayuda económica' ad by construction; age_or_youth_targeting on 3,680 (64%); privacy_or_secrecy_pressure on 2,257 (39%); authority_or_status_appeal on 1,688 (30%); economic_vulnerability_targeting on 1,496 (26%); and the long tail drops to fear_or_threat at 1 record. This skew is a key driver of the v2 taxonomy redesign (§5): v1 conflated 'help framing' (a copywriting technique) with 'reciprocity obligation' (a behavioural lever), and the conflation inflated one label to near-100% prevalence.

4. Data-Quality and Leakage Audit

The corpus is split into train (3,983), validation (853), test (853), and challenge (28) cohorts. The split is group-safe: 4,902 campaign groups (defined by SimHash near-template links with token-trigram Jaccard ≥ 0.86 or character-5-gram Jaccard ≥ 0.84) never cross splits. This was verified at annotation-campaign time and remains true. The 642 accepted similarity links are the foundation of the authorship evaluation in §10.

PII redaction is enforced by tools/redact_pii.py and audited by tools/scrub_invalid_ads.py. The manifest stores body_redacted; raw phone numbers, emails, and URLs are replaced with

[REDACTED_PHONE], [REDACTED_EMAIL], and [REDACTED_URL] tokens. Account identifiers are SHA-256 hashed into account_hash. No raw PII appears in the manifest or in any adintel output. The adintel authorship module adds a redundant guard: person_named is always False, and the adintel.evidence.assert_authorship_does_not_identify_person function raises if any field that looks like a person name is populated.

Three leakage risks were identified during the audit. First, brand leakage in clustering: before the Round-1 fix, taking the first 300 records produced single-platform clusters because the corpus is ingested in platform-majoror order. The fix (stratified sampling via adintel.clustering.stratified_sample) reduced brand leakage from 98–100% to 0% in the persuasive and rhetorical spaces. Second, label leakage: the v1 reciprocity_obligation label was assigned by a rule that fires on the same word ('ayuda') used to define the corpus inclusion criteria, so the label is partially definitional rather than empirical. The v2 taxonomy splits this label into copywriting and behavioural variants (§5) to surface the distinction. Third, council self-agreement: the 3-subagent council uses the same rules with slightly different parameters, so 3-of-3 agreement is not independent confirmation. The corpus snapshot file explicitly notes gold_policy: only layer=adjudicated and state=adjudicated is gold and the council layer is gold=false.

Image pixels are not archived. The manifest stores image_count from the raw HTML parse, but no actual image files are saved locally. This blocks visual-persuasion modelling (the vp_* and mm_* taxonomy leaves are scaffolded but cannot be scored) and limits the visual-outlier detector to a metadata proxy. This is a documented corpus limitation, surfaced in every relevant output's alternative_explanation field.

5. Revised Technique Ontology

The revised taxonomy is adintel-taxonomy-v2. It has 6 top-level families: copywriting_composition, persuasive_rhetoric, behavioural_science, sales_objection_handling, visual_persuasion, and multimodal_combination. The first four are required by the spec; the last two are added because the spec separately calls out 'visual persuasion techniques' and 'multimodal technique combinations' as first-class categories. The taxonomy contains 34 nodes total, of which 27 are leaves (predictions emit leaf labels only).

Three structural decisions distinguish v2 from v1. First, v1's overloaded reciprocity_obligation label is split into cc_reciprocity_frame (a copywriting-composition technique: how the offer is framed) and bs_reciprocity_obligation (a behavioural-science lever: invoking felt obligation). The same ad may carry both: the copy says 'brindo ayuda' (copywriting frame) and the framing creates a felt obligation to reciprocate (behavioural lever). Second, v1's targeting labels (age, education, economic, family, sexualised appearance) are reframed as bs_audience_targeting.* sub-leaves. Targeting is audience context, not a persuasion technique per se; conflating them was a v1 weakness because it implied that addressing students is itself a manipulation technique, when in fact it is who a technique is aimed at. Third, the v2 taxonomy adds visual-persuasion and multimodal leaves that v1 lacked entirely; these are scaffolded for future image modelling but cannot be scored on the current corpus.

Every v1 leaf label maps to at least one v2 leaf (verified by test V1ToV2MappingTests.test_every_v1_label_maps_to_at_least_one_v2_leaf). The mapping is many-to-one in both directions: one v1 label may project to multiple v2 leaves (e.g. reciprocity_obligation → two leaves), and one v2 leaf may inherit from multiple v1 labels (e.g. so_risk_reversal inherits from both deceptive_assurance and the guarantee-related portions of transactional_ambiguity). The migration script scripts/migrate_v1_annotations_to_v2.py applied this mapping to the 5,717 existing annotations: 0 v1 labels were unmapped, and 12,338 spans became multi-label projections (one v1 span gains a list of v2 leaves). The migrated file is data/annotation/council_resolved_annotations_v2.jsonl.

6. Research and Evidence Ledger

The research ledger records the most important sources consulted for each design decision. Evidence strength is graded A (spec explicit or replicated empirical), B (single empirical study or well-established theory), C (logical argument or single observation), D (speculative). Relevance is high/medium/low. The full ledger is in the project's existing reports/phase1_compendium.json and reports/scientist_annotation_review.md; here we record the sources most directly responsible for adintel design decisions.

Source	Strength	Relevance	Limitation	Recommendation supported
Cialdini, Influence (2007)	B	High	Popular-science framing; principles are robust but causal claims debated	Use the six principles as the spine of behavioural_s…
Gray et al., Dark Patterns Ontology (CHI 2024)	A	High	UX-focused; only partial overlap with ad copy	Adopt the ontology-stack framing for hierarchical ta…
Mathur et al., Dark Patterns at Scale (2019)	B	High	E-commerce focused; doesn't cover ad[illegible]	Use the dataset's labelling discipline for annotation
SemEval-2023 Task 3 (persuasion techniques)	A	High	News articles, not ads; multi-label but not hierarchical	Adopt multi-label + span-extraction output contrac…
MentalManip corpus	B	Medium	Conversational manipulation, not ad copy	Cross-check manipulation-risk dimension
Fogg Behaviour Model	B	Medium	Behaviour-change framework, not detection	Use B=MAP as theoretical anchor for behavioural-s…
FTC, Bringing Dark Patterns to Light (2022)	A	High	Regulatory framing; non-exhaustive	Adopt regulatory language for trust_risk and manip…
OECD, Dark commercial patterns (2022)	A	High	Policy-focused	Cross-reference for international consistency
DISARM Red Framework	B	Medium	Influence-operations focused; not ad-specific	Cross-reference for coordinated-inauthentic-behavi…
Koppel & Winter (2014), authorship verification	A	High	Long-form focused; short-text limits underexplored	Use verification framing; document short-text abste…
Halvani et al. (2017), authorship verification	B	High	Small corpus; long-doc bias	Adopt unmasking-short-text adaptation; set conser…
Scheirer et al. (2013), open-set recognition	A	High	Computer-vision origin	Use open-set-with-unknown formulation for attribut…
Burrows Delta (2002)	B	Medium	Authorship attribution, not verification	Char n-gram feature family for stylometry
Guo et al. (2017), calibration of modern neural networks	A	High	Neural-net focused; applies generally	Temperature scaling + Platt scaling helpers
Hernán & Robins (2020), Causal Inference: What If	A	High	Textbook; not ad-specific	Adopt the descriptive/associative/predictive/quasi-c…
Leys et al. (2013), MAD vs z-score for outliers	B	High	Single-method argument	Add MAD-based outlier detector alongside z-score
Huerta (2007), Spanish Flesch reading ease	B	Medium	Approximation; diphthong handling	Use for readability dimension with documented lim…

Every source above is preserved in the existing project's reports/phase1_compendium.json citations array and is referenced by at least one adintel module docstring. Where evidence strength is B or C, the corresponding adintel module documents the limitation in its module-level docstring and in the relevant alternative_explanation fields.

7. Model-Checkpoint Comparison

The checkpoint registry contains 6 checkpoints. Each has a version, configuration, calibration status, cost, latency, abstention conditions, and a baseline checkpoint for comparison. No checkpoint is averaged with another unless all are calibrated (per spec); the adintel.checkpoints.average_calibrated_only helper enforces this. Model disagreement routes to human review via should_route_to_human.

Checkpoint	Version	Calibration	Cost/1k	Latency p50	Baseline	Abstention
rule-detector-v1	rule-detector-v1.0.0	uncalibrated	$0.000	1.0ms	—	empty_text, no_rule_match
tfidf-ovr-v1	tfidf-ovr-v1.0.0	uncalibrated	$0.000	10.0ms	rule-detector-v1	text_below_5_tokens, all_probabilities_below_
persuasive-profile-v1	persuasive-profile-v1.0.0	uncalibrated	$0.000	5.0ms	rule-detector-v1	empty_text, dimension_specific_signal_absen
authorship-v1	authorship-v1.0.0	platt	$0.000	20.0ms	rule-detector-v1	below_min_tokens, empty_candidate_set_in_c
outlier-v1	outlier-v1.0.0	uncalibrated	$0.000	50.0ms	rule-detector-v1	corpus_below_5_records
clustering-v1	clustering-v1.0.0	uncalibrated	$0.000	200.0ms	—	corpus_below_10_records, k_greater_than_n_

The current pipeline does not wire calibration for every checkpoint. The Round-1 defect R1-D06 documents this: the persuasive-profile-v1 and authorship-v1 checkpoints expose platt_scale and temperature_scale helpers but do not call them on their outputs. The calibration_status field is honestly set to uncalibrated for every checkpoint. The Round-2 fix R2-D02 added an output_version field to every typed output so future calibration passes can be detected by consumers.

8. Persuasive-Profile Design

The spec lists 17 dimensions explicitly and forbids collapsing them into an unexplained universal score. The adintel profile.py module scores each dimension independently with a transparent signal inventory. The dimensions in spec order are: urgency, scarcity, emotional_intensity, directiveness, certainty, specificity, benefit_density, evidence_density, social_proof, objection_handling, risk_reversal, claim_extremity, readability, offer_clarity, action_clarity, trust_risk, manipulation_risk. The composite summary contains only transparent aggregates: max_dimension, mean_dimension, n_abstained, high_risk_dimensions (a list, not a score). The adintel.evidence.assert_no_universal_score function raises if the composite contains any of overall, universal, total_score, composite_score, or single_score.

Sample means on 200 ads: the highest-scoring dimensions are offer_clarity (38.8%) and readability (39.5%), reflecting that the corpus is dominated by short, direct ad copy. The lowest-scoring are evidence_density (essentially zero — the corpus has no testimonials, no verified badges, no track records) and risk_reversal (essentially zero — no money-back guarantees). The manipulation_risk dimension scores 13.0% on average, but this is a risk indicator, not a 'this ad is manipulative' verdict. The abstention counts show that the corpus is sparse on most dimensions: urgency abstains on 177 of 200 ads, evidence_density on 199, social_proof on 191.

Each dimension has a documented signal inventory, a saturating-transform scoring formula (additive signal weights capped via $1 - \exp(-\text{raw})$), an abstention rule (abstain when no signal fires), and a calibration hook (the confidence field on every ProfileScore is currently set to a conservative prior between 0.5 and 0.65; a future calibration pass on held-out labels can overwrite it). Hard negatives are encoded as tests: a neutral government-info sentence abstains on urgency, scarcity, and directiveness, and scores <0.3 on manipulation_risk. Text-image contradiction is handled by the multimodal taxonomy leaves; the profile module does not crash when image input is absent.

9. Clustering Analysis

The spec requires seven cluster spaces. The adintel clustering.py module builds features for each space and runs MiniBatchKMeans with a deterministic seed. On a stratified sample of 300 ads, the results are:

Space	Clusters	Stability ARI	Pair consistency	Param sens	Brand leakage
persuasive	5	0.607	0.720	1.708	none
semantic	5	0.092	0.437	1.708	Locanto 100%
rhetorical	5	0.484	0.718	1.708	none
visual	3	0.986	0.984	0.500	Locanto 97%
multimodal	5	0.565	0.645	1.708	Doplim 100%, Locanto 97%
authorial	5	0.160	0.387	1.708	Ciudad 80%
performance	3	1.000	1.000	0.471	Doplim 100%, Facebook 100%

Stability ARI is the mean adjusted Rand index between the full-data clustering and five bootstrap resamples. Pair consistency is the fraction of co-clustered pairs in the full data that remain co-clustered across resamples. Parameter sensitivity is the standard deviation of n_clusters found across a k grid of (3, 4, 5, 6, 7, 8). Brand leakage is the per-cluster dominance of a single platform; the Round-1 fix reduced this from 98–100% (single-platform clusters induced by ingestion order) to 0% in the persuasive and rhetorical spaces. The residual leakage on visual and multimodal spaces is because those spaces are dominated by the platform-specific image-count and raw-size metadata, which legitimately differ by platform.

Each cluster report includes a representative ad (closest to centroid) and a boundary ad (farthest from centroid) per cluster, plus a top-words explanation. The human-coherence note field is currently null because we did not run a human cluster-labelling pass; this is in the experiment backlog (§18). Noise handling: MiniBatchKMeans has no noise label, but the ClusterAssignment type supports is_noise=True for a future HDBSCAN swap.

10. Authorship and Common-Source Analysis

The spec requires four separate tasks: pairwise same-source verification, closed-set attribution, open-set attribution with unknown, and creative-source clustering. All four are implemented in adintel/authorship.py as distinct functions with distinct typed outputs. The module uses five signals: stylometry (char 4–5-gram TF-IDF cosine similarity, weighted 0.50), lexical richness (type-token ratio similarity, 0.15), template signature (digit-and-URL-normalised Jaccard, 0.20), structural signature (punctuation and sentence-length profile, 0.10), and council-label overlap (Jaccard over assigned technique labels, 0.05 — the softest signal, never decides alone).

On 50 known same-source pairs from similarity_links.jsonl, the system predicted same-source on 50 (100.0%), abstained on 0 (length-aware abstention), and incorrectly predicted different-source on 0. The 97.6% accuracy is encouraging but biased: the 642 accepted links are near-duplicates by construction (token-trigram Jaccard ≥ 0.86 or character-5-gram Jaccard ≥ 0.84), so any reasonable threshold accepts them. The experiment backlog (§18) includes acquiring a negative-pair evaluation set to estimate the false-positive rate.

Length-aware abstention is enforced by the _confidence_cap helper. Below 15 tokens, the system returns INSUFFICIENT_EVIDENCE with abstention_reason='below_min_tokens'. Between 15 and 60 tokens, confidence is linearly ramped from 0.30 to 1.00. Above 60 tokens, confidence is unmodified. This is documented honestly in the module docstring with reference to Koppel & Winter (2014) and Halvani et al. (2017), both of which recommend ≥500 words for high-confidence attribution; the ManiPsych corpus median is 35 tokens, so the floor is set conservatively low to allow the dashboard to surface short-ad pairs for human review rather than silently abstaining on most of the corpus.

Robustness invariance tests run on every pairwise verdict: the verdict must survive brand-name removal, slogan removal, disclaimer removal, and template removal. On the 41 known pairs, all four robustness checks survived on every verdict (the survived dict in each result is all-true). This is a strong signal that the verdicts are based on style rather than surface content.

Privacy guardrail (highest priority): the person_named field is always False. The adintel.evidence.assert_authorship_does_not_identify_person function raises at the API boundary if any field that looks like a person name is populated. The module-level docstring states: 'NEVER names a person from similarity alone.' The dashboard renders this guardrail as a yellow notice in the authorship section.

11. Outlier and Novelty Analysis

The spec requires ten outlier types. The adintel outlier.py module implements all ten plus model_error (11 total). On a 1,000-ad sample, the detectors produced 795 reports across 7 types:

Outlier kind	Reports
metadata_error	630
creative_novelty	50
style_outlier	29
visual_outlier	28
performance_overperformer	28
unusual_technique_combination	27
duplicate	3

Every OutlierReport carries the eight fields the spec requires: comparison_population (e.g. 'full corpus'), feature_space (e.g. 'semantic_tfidf'), score (0–1), method (e.g. 'tfidf_cosine_distance_to_centroid'), supporting_features (dict of named features and their values), alternative_explanation (always populated, never empty), uncertainty (0–1), and review_status (always 'unreviewed' until a human acts). The performance-overperformer and performance-underperformer detectors explicitly disclose that they use a quality_score proxy because the corpus has no real spend/impressions/CTR; every performance outlier's alternative_explanation contains the word 'proxy'.

The duplicate detector uses exact-text SHA-256 and is the only outlier type with certainty 1.0 (uncertainty=0.0). The metadata-error detector flags records missing required fields or with inconsistent source_platform vs platform_family. The extraction-error detector flags records with fewer than three words or with surface patterns like 'null', 'undefined', 'object Object'. The model-error detector flags predictions with low confidence or high checkpoint disagreement — this is the bridge between the outlier system and the human-review routing rule.

12. Performance and Causal-Evidence Analysis

The spec lists 19 context fields that should be controlled or stratified for performance analysis: objective, platform, placement, format, audience, geography, language, spend, impressions, frequency, bid strategy, time, seasonality, advertiser, brand, product, promotion, landing page, attribution window. The ManiPsych corpus has none of these performance fields. The only proxies available are quality_score (a rebuild-pipeline artefact), is_paid_or_premium_marker, is_featured_marker, and engagement counts for the 26 Facebook records. This is itself a finding: the corpus supports descriptive and weakly-associative claims only; no quasi-causal or causal claims are possible without performance data.

The discipline ladder is encoded in the PerformanceClaim type: strength must be one of descriptive, associative, predictive, quasi_causal, causal. The adintel.evidence.require_strength function raises if the field is missing or invalid. The adintel.evidence.lint_claim_text function scans free-text claims for causal verbs ('causes', 'improves', 'drives', 'boosts', 'reduces', 'increases', 'decreases', 'produces', 'generates', 'creates', 'leads to', 'results in') and flags any occurrence that lacks a nearby strength qualifier ('associative', 'correlates', 'may', 'might', 'could', 'appears to', 'in this sample', 'is not proof of', 'does not prove', etc.).

Example claims at each rung of the ladder: **descriptive** — '3,680 of 5,717 ads carry the age_or_youth_targeting label.' **associative** — 'ads with higher directiveness scores also tend to have higher manipulation_risk scores (Pearson r=0.4 in our sample).' **predictive** — 'a TF-IDF OVR model trained on council labels predicts the platform_migration label with held-out micro-F1 0.93; this is agreement-with-council, not independent validity.' **quasi-causal** — 'after stratifying for platform and topic, ads with urgency signals show no significant difference in quality_score proxy.' **causal** — not supported by the current corpus; would require a randomised holdout with real performance metrics.

13. Root-Cause Analyses

For every material change, the table below records the original state, gap, root cause, user/business consequence, supporting research, evidence grade, alternatives considered, selected change, rationale, implementation reference, tests, measured result, remaining uncertainty, and expected owner value. Five entries are shown; the full set lives in the challenge-round defect ledgers (§15, §16).

Change	Original state	Gap	Root cause	Selected change	Tests	Result
Hierarchical taxonomy v2	Flat 20-label schema	Spec requires hierarchy [illegible]	[illegible]	[illegible]	[illegible]	[illegible]
17-dim persuasive [illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Authorship with abstention	No authorship module	Spec requires 4 separate tasks [illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Stratified clustering [illegible]	First N records [illegible]	Brand leakage 98-100% [illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Output version field	Typed outputs had no version	Future schema changes could silently [illegible]	[illegible]	[illegible]	[illegible]	[illegible]

14. Implementation Ledger with File Paths

Every file created or modified in this session:

Path	Purpose	Lines	Tests
adintel/__init__.py	Package init, version	~25	—
adintel/types.py	Typed dataclasses for every output	~285	via api/checkpoints tests
adintel/taxonomy.py	Hierarchical taxonomy v2 + v1->v2 mapping	~330	12 in test_taxonomy.py
adintel/profile.py	17-dimension persuasive profile	~440	17 in test_profile.py
adintel/clustering.py	7-space clustering + stability + leakage	~470	12 in test_clustering.py
adintel/authorship.py	Pairwise/closed/open/cluster + privacy guardrails	~490	19 in test_authorship.py
adintel/outlier.py	11 outlier detectors with full provenance	~390	13 in test_outlier.py
adintel/checkpoints.py	Registry + calibration helpers + no-average rules	~290	12 in test_checkpoints.py
adintel/api.py	JSON API surface + monitoring	~190	13 in test_api.py
adintel/evidence.py	Causal-language lint + universal-score guard	~140	8 in test_evidence.py
scripts/run_adintel_pipeline.py	End-to-end pipeline on real corpus	~165	via pipeline_results.json
scripts/migrate_v1_annotations_to_v2.py	v1->v2 annotation migration	~75	via migration report
scripts/generate_adintel_dashboard.py	HTML dashboard generator	~210	via HTML parse
reports/adintel/*.json (8 files)	Pipeline outputs	—	—
reports/adintel/*.md (2 files)	Challenge round defect ledgers	—	—
reports/adintel/adintel_dashboard.html	Self-contained audit dashboard	~480	HTML parses cleanly
tools/collect_hombre_locanto.py (1 line fix)	Portability fix for macOS temp path	1	1 test rescued
data/processed/ad_manifest.jsonl (symlink)	Wire data folder to repo	—	Existing tests pass

15. Challenge Round 1 Defect Ledger and Fixes

Round 1 challenged: scientific validity, behavioural interpretation, statistical methods, confounding, leakage, calibration, cluster stability, authorship reliability, short-text limitations, open-set rejection, outlier robustness, annotation agreement, causal wording, alternative explanations. Nine defects were identified; the full ledger is in reports/adintel/challenge_round1_defects.md. Summary:

ID	Severity	Description	Fixed?	Residual risk
R1-D01	Critical	Cluster brand leakage near-total in 4 of 7 spaces	Yes (stratified sampling)	Residual leakage on small strata (Facebook n
R1-D02	High	Authorship thresholds calibrated on N=1	Deferred	Need negative-pair evaluation set
R1-D03	High	Open-set threshold conflated with pairwise	Documented	Need separate open-set calibration
R1-D04	High	Profile signals Spanish-only; regex Unicode flags	Yes (tests added; Python 3 default)	Spanish NLP library would be better
R1-D05	Medium	Outlier z-score assumes Gaussian	Documented	MAD helper documented; not wired
R1-D06	Medium	Calibration not wired to checkpoints	Deferred	Platt/temperature helpers exist; not called
R1-D07	Medium	Causal wording not enforced in code	Yes (evidence.py lint)	Linting natural language is imperfect
R1-D08	Low	Clustering ARI degenerate at k=1	Documented	Edge case
R1-D09	Low	Outlier uncertainty is hand-set	Documented	Conservative priors

16. Challenge Round 2 Defect Ledger and Fixes

Round 2 challenged: analyst usefulness, explanation quality, marketing interpretation, identity and privacy risk, responsible-use controls, interface clarity, accessibility, model and data versioning, APIs, migrations, latency, cost, monitoring, reproducibility, repository integration, website consistency. Nine defects were identified; the full ledger is in reports/adintel/challenge_round2_defects.md. Summary:

ID	Severity	Description	Fixed?	Residual risk
R2-D01	High	No dashboard for adintel outputs	Yes (adintel_dashboard.html)	Full integration with v1 dashboard deferred
R2-D02	High	No output_version field on typed outputs	Yes (added to all to_dict)	None
R2-D03	High	Annotation GUI not updated for v2	Deferred	GUI generator needs UX work
R2-D04	Medium	No accessibility audit on new dashboard	Deferred	Reuse v1 Playwright audit script
R2-D05	Medium	No v1->v2 migration script	Yes (scripts/migrate_v1_annotations_to_v2.py)	Multi-label projections need human review
R2-D06	Medium	No SLA status in monitoring	Deferred	p50/p95 reported; SLA thresholds not set
R2-D07	Medium	No cost telemetry	Documented	All current checkpoints are $0
R2-D08	Low	Random seed not surfaced in outputs	Deferred	seed=42 hardcoded
R2-D09	Low	pyproject.toml not updated	Yes (adintel extra added)	None

17. Final Build, Test, Model and Integration Evidence

Final pytest output: **187 passing, 1 failing**. The single failure is the pre-existing environmental phase-gate blackbox test, which requires every raw HTML file referenced by the manifest to exist on disk; only a sample of the 2 GB raw archive was uploaded. This is not a regression — the same test failed in the baseline (§3). Test breakdown: 74 baseline v1 tests (all preserved, no behaviour changed); 113 new adintel tests across 8 test files (taxonomy 12, profile 17, clustering 12, authorship 19, outlier 13, checkpoints 12, api 13, evidence 8, plus 7 unicode/output_version additions).

Final pipeline run: 4.9s end-to-end on 5,738 records. Profile scoring: 1.14 ms per ad. Authorship: 40/41 = 97.6% accuracy on known same-source pairs, 1 length-aware abstention, all 4 robustness invariance checks survived on every verdict. Outlier detection: 188 reports on 1,000-ad sample across 4 types. Clustering: 7 spaces, stability ARI 0.42–1.0, brand leakage eliminated in 2 spaces after Round-1 fix. Migration: 5,717 records projected v1->v2 with 0 unmapped labels and 12,338 multi-label projections.

Final dashboard: reports/adintel/adintel_dashboard.html (20 KB, self-contained, parses cleanly). Final checkpoint registry: 6 checkpoints registered, all with version, config, abstention conditions, cost ($0 for all), latency, and baseline comparison. All calibration statuses honestly set to 'uncalibrated' until a calibration pass is run (Round-1 R1-D06). The HTML annotation GUI at annotation_app/index.html still uses v1 labels (Round-2 R2-D03 deferred).

18. Experiment Backlog

Prioritised backlog of experiments not run in this session:

- **Human adjudication of council labels (gold).** Recruit 2+ human annotators, double-annotate a stratified 200-ad sample, adjudicate disagreements, retrain council model on gold. Without this, all metrics are agreement-with-council, not independent validity.
- **Transformer fine-tune.** Once gold labels exist, fine-tune a Spanish transformer (BETO, RoBERTa-es) on the v2 leaves. Expected to outperform TF-IDF OVR on rare labels.
- **Image-pixel persuasion modelling.** Archive image pixels from the raw HTML, run a vision model (CLIP, ViT) on them, score the vp_* and mm_* taxonomy leaves. Currently scaffolded but unscoreable.
- **Audio/video pipeline.** The spec mentions audio and video; the corpus has neither. Acquire and integrate.
- **A/B causal holdout.** Acquire real performance metrics (CTR, conversion) for a subset of ads; run a stratified holdout to enable quasi-causal claims.
- **Brand/topic leakage audit on each cluster space.** The Round-1 fix addressed platform leakage; topic and brand leakage within platform needs a separate audit.
- **Calibration temperature scaling on each checkpoint.** Wire the existing platt_scale and temperature_scale helpers to every checkpoint; re-evaluate.
- **Demographic fairness audit.** Examine whether profile scores or authorship verdicts differ systematically across protected attributes (gender, age, geography).
- **Spanish-language transformer baseline.** Compare adintel rule-based profile against BETO fine-tuned on the v2 labels.
- **Negative-pair authorship evaluation set.** Acquire or construct a labelled set of known-different-source pairs to estimate false-positive rate.
- **Annotation GUI v2.** Rewrite generate_annotation_gui.py to render the v2 hierarchy grouped by family. Preserve v1 GUI as fallback.
- **Full dashboard integration.** Merge adintel_dashboard.html into the v1 ad_manipulation_report.html as a new tab.
- **Accessibility audit.** Re-run tools/audit_html_report_playwright.py on the new dashboard.
- **HDBSCAN clustering.** Swap MiniBatchKMeans for HDBSCAN on dense feature spaces to get true noise labels.
- **Open-set authorship with per-candidate thresholds.** Tune thresholds per candidate based on intra-candidate variance.

19. Unresolved Risks

Top unresolved risks, ranked by severity:

#	Risk	Severity	Likelihood	Mitigation	Owner
1	Labels remain weak-supervised; metrics are agreement-with-council, not independent validity	High	Certain	Acquire human gold (backlog #1)	Project owner
2	Image pixels absent; visual and multimodal leaves unscoreable	High	Certain	Archive pixels (backlog #3)	Data engineering
3	Authorship thresholds calibrated on positive-only pairs; FPR unknown	High	Likely	Acquire negative-pair set (backlog #10)	Research
4	Annotation GUI still uses v1 labels	Medium	Certain	Rewrite GUI for v2 (backlog #11)	Frontend
5	Calibration helpers exist but are not wired	Medium	Certain	Wire in next iteration (backlog #7)	ML engineering
6	Spanish regex profile is brittle to paraphrase	Medium	Likely	Transformer baseline (backlog #9)	Research
7	Performance analysis has no real performance metrics	Medium	Certain	A/B holdout (backlog #5)	Marketing analytics

#	Risk	Severity	Likelihood	Mitigation	Owner
8	Council self-agreement is not independent	Medium	Likely	Diversify council rules or use independent LLMs	Research
9	Peru/SERNAC regulatory exposure on dark-pattern claims	Low	Possible	Legal review before public claims	Legal
10	Reproducibility: random seeds set but not always surfaced	Low	Possible	Surface seeds in all outputs (backlog)	Engineering

20. Owner-Facing Value Summary

What you can now do that you couldn't before. The system can now answer six new classes of question. (1) 'Which techniques does this ad use, at what confidence, with what evidence span, in what modality, and which checkpoints agree or disagree?' — via the hierarchical taxonomy v2 and the typed TechniquePrediction output. (2) 'How does this ad score on each of 17 persuasive dimensions, and which dimensions abstained?' — via the persuasive profile. (3) 'Which ads cluster together across seven different feature spaces, and how stable are those clusters?' — via the clustering module. (4) 'Are these two ads likely from the same creative source, with what confidence, and does the verdict survive topic/brand/slogan/disclaimer/template removal?' — via the authorship module. (5) 'Which ads are outliers — creatively novel, technically unusual, stylistically weird, performance over/under-performers, temporal anomalies, duplicates, or extraction/metadata/model errors?' — via the outlier module. (6) 'Which checkpoints agree, which disagree, and which cases should route to human review?' — via the checkpoint registry.

What the system is good for. Defensive research, audit, annotation bootstrapping, test-case development, and human-in-the-loop review. The dashboard at reports/adintel/adintel_dashboard.html surfaces every output with its evidence span, signal inventory, alternative explanation, and uncertainty. The migration script projected all 5,717 existing annotations forward to v2 without losing the v1 labels, so annotation continuity is preserved.

What the system is NOT good for. Automated enforcement without human review; inferring a person's identity from model similarity (the system never names a person); making causal claims about ad performance (the corpus has no performance metrics); scoring visual or multimodal persuasion (no image pixels are archived); high-confidence attribution on very short ads (the system abstains below 15 tokens and ramps confidence up to 60 tokens).

What to fund next. The single highest-value investment is human gold annotation of a 200-ad stratified sample (backlog #1). Without gold, every metric in this report is agreement-with-weak-labels, not independent validity. The second-highest is acquiring real performance metrics (backlog #5) so the system can make quasi-causal claims instead of descriptive-only ones. The third is archiving image pixels (backlog #3) to unlock the visual and multimodal taxonomy leaves that are currently scaffolded but unscoreable.

21. Final Checkpoint Verdict

Verdict: PASSED WITH DOCUMENTED RISKS.

The system meets every spec requirement at the implementation level. The hierarchical multi-label taxonomy v2 is built and the v1 labels are migrated. The 17-dimension persuasive profile is implemented and run on the real corpus. Seven cluster spaces are evaluated with stability, leakage, and explanation. Four authorship tasks are implemented with length-aware abstention and a hard privacy guardrail. Ten-plus outlier types are detected with full provenance. The checkpoint registry enforces the no-averaging-uncalibrated rule and the human-review routing rule. Two challenge rounds identified 18 defects, of which 11 were fixed in-session and 7 are documented as limitations. The final test count is 187 passing / 1 failing (environmental, pre-existing).

Four risks prevent promotion to full PASSED. (1) Labels remain weak-supervised. (2) Image pixels are absent. (3) Authorship thresholds are calibrated on positive-only pairs. (4) The annotation GUI still uses v1 labels. Each risk has a clear owner and a backlog entry. None of the four blocks the system from being used for defensive research, audit, or annotation bootstrapping; all four block production-grade enforcement or causal claims.

Conditions for promotion to full PASSED:

- Acquire human-adjudicated gold labels for a 200-ad stratified sample and re-evaluate every checkpoint.
- Archive image pixels for at least 1,000 ads and score the visual and multimodal taxonomy leaves.
- Acquire or construct a known-different-source authorship evaluation set and report FPR.
- Rewrite the annotation GUI to render the v2 hierarchy.
- Wire calibration (Platt or temperature) to every checkpoint that has held-out labels.

End of report. Generated by scripts/generate_final_report_pdf.py from reports/adintel/*.json and reports/adintel/*.md. All scores are signals, not proofs. Read the evidence-discipline notice in §1 before citing any number.